

To organize and make these comprehensive results easy to compare, here is the complete breakdown of all reported metrics from each study compiled into structured tables.

Study 1: Salim & Ibrahim (2026) — 30-Day Diabetes Readmissions

This study evaluated several models on the **Diabetes 130-US Hospitals dataset** (101,766 encounters) 1, 2.

Baseline Model Comparison (Nested Cross-Validation)

Model,Nested Cross-Validation ROC-AUC 3
Stacking Ensemble (Logistic Regression + XGBoost + LightGBM base models),0.665
XGBoost (Selected Primary Model),0.664
LightGBM,0.660
Logistic Regression,0.657
Random Forest,0.650

Calibrated XGBoost Performance Metrics (Full-Dataset)

Metric,Value 3
Full-Dataset AUC (Post-Calibration Retraining),0.688
Brier Score,0.094
Precision-Recall AUC (PR-AUC),0.215
F1-Score (Minority Readmitted Class),0.27
Sensitivity (Recall) (at 10% risk threshold),0.723
Specificity (at 10% risk threshold),0.530
Positive Predictive Value (PPV) (at 10% risk threshold),0.162
Negative Predictive Value (NPV) (at 10% risk threshold),0.938
"Predicted High Risk Patients per 1,000",497.9
"True Readmissions Identified per 1,000",80.6

XGBoost Performance Across Programmatic Decision Thresholds

Decision Threshold,Sensitivity 4,Specificity 4,PPV 4,NPV 4,"High Risk per 1,000 4"
0.05,1.000,0.073,0.119,1.000,934.8
0.10,0.881,0.524,0.189,0.972,521.0
0.15,0.570,0.839,0.308,0.940,206.5
0.20,0.264,0.967,0.501,0.913,58.9
0.25,0.108,0.994,0.707,0.899,17.1
0.30,0.038,0.999,0.886,0.892,4.8

Study 2: Ghosh et al. (2026) — Restricted vs. Comprehensive EHR Code Sets

This retrospective comparative study analyzed **50,000 encounters** from the New York State Emergency Department Database, comparing a **Limited Feature Set** (11 features) to a **Rich Feature Set** (up to 135 features) 5.

Model Architecture, Feature Set, AUROC (95% CI) 6, F1-Score (95% CI) 6, Accuracy (95% CI) 6, MCC (95% CI) 6

Random Forest, Limited, 0.5596 (0.5440–0.5739), 0.2657 (0.2469–0.2829), 0.7054 (0.6964–0.7137), 0.0823 (0.0610–0.1024)

, Rich, 0.5541 (0.5393–0.5671), 0.2528 (0.2351–0.2699), 0.7110 (0.7019–0.7192), 0.0761 (0.0536–0.0975)

CatBoost, Limited, 0.5369 (0.5236–0.5518), 0.2555 (0.2381–0.2723), 0.6795 (0.6698–0.6888), 0.0514 (0.0312–0.0713)

, Rich, 0.5333 (0.5191–0.5470), 0.2509 (0.2343–0.2682), 0.6752 (0.6665–0.6842), 0.0438 (0.0247–0.0633)

Multilayer Perceptron (MLP), Limited, 0.5386 (0.5232–0.5528), 0.2860 (0.2695–0.3023), 0.6550 (0.6450–0.6643), 0.0641 (0.0422–0.0846)

, Rich, 0.5287 (0.5156–0.5430), 0.2205 (0.2032–0.2377), 0.7116 (0.7034–0.7205), 0.0487 (0.0283–0.0705)

DistilBERT, Limited, 0.5542 (0.5411–0.5666), 0.3434 (0.3303–0.3570), 0.4408 (0.4312–0.4515), 0.0663 (0.0472–0.0856)

, Rich, 0.5703 (0.5565–0.5842), 0.3406 (0.3275–0.3543), 0.4893 (0.4796–0.4987), 0.0723 (0.0530–0.0910)

Study 3: Pandey et al. (2025) — Multimodal Hybrid Modeling with ClinicalT5

This study evaluated hybrid modeling (ClinicalT5 200-word clinical note chunk embeddings + structured EHR features) using a **79,731 patient cohort from MIMIC-IV 7-9**.

Model Configuration, Accuracy 10, Recall 10, Precision 10, Specificity 10, F1-Score 10, AUROC 10, MCC 10

ClinicalT5 + LGBM, 0.63, 0.66, 0.63, 0.60, 0.64, 0.68, 0.26

ClinicalT5 + VotingClassifier, 0.63, 0.68, 0.63, 0.58, 0.65, 0.68, 0.26

ClinicalT5 + XGBClassifier, 0.62, 0.66, 0.62, 0.58, 0.64, 0.66, 0.24

PubMedBERT (Text-Only Baseline), 0.60, 0.69, 0.60, 0.50, 0.64, 0.64, 0.20

ClinicalT5 + Neural Network, 0.58, 0.76, 0.56, 0.40, 0.65, 0.62, 0.17

Study 4: Almeida et al. (2025) — GraphSAGE GNN with Clinical Notes

This study integrated BioClinicalBERT embeddings of discharge clinical notes and structured EHR features as a graph network representing **303,571 hospital admissions in MIMIC-IV 11-13**.

GraphSAGE Node Feature Ablation Performance (Sorted by AUROC)

Active Node Features, AUROC 14, Balanced Accuracy 14

"All Features (Admissions, Procedures, Diagnoses, Lab Events, Clinical Notes)", 0.727, 0.667

"Admissions, Procedures, Diagnoses, Clinical Notes (No Lab Events)", 0.719, 0.660

"Admissions, Diagnoses, Lab Events, Clinical Notes (No Procedures)", 0.707, 0.650

"Admissions, Procedures, Lab Events, Clinical Notes (No Diagnoses)", 0.705, 0.650

"Admissions, Procedures, Diagnoses, Lab Events (No Clinical Notes)",0.704,0.649

"Admissions, Lab Events, Clinical Notes",0.699,0.645

"Admissions, Lab Events (No Clinical Notes, Diagnoses, or Procedures)",0.691,0.639

GraphSAGE Baseline Architecture Comparison (20-Fold Cross-Validation)

Model,AUROC 15,Balanced Accuracy 15

GraphSAGE (Proposed GNN),0.7163,0.6585

Multilayer Perceptron (MLP),0.7089,0.6501

Logistic Regression (LR),0.6729,0.6298

 Since you are comparing clinical and modeling details across these studies, would you like me to generate a structured audio overview/podcast script discussing how structured clinical data vs. free-text notes affect model performance?